



Sky: SparkLend Advance

Security Review

Cantina Managed review by:

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1 Introduction

1.1 About Cantina

Cantina is a security services marketplace that connects top security researchers and solutions with clients. Learn more at cantina.xyz

1.2 Disclaimer

Cantina Managed provides a detailed evaluation of the security posture of the code at a particular moment based on the information available at the time of the review. While Cantina Managed endeavors to identify and disclose all potential security issues, it cannot guarantee that every vulnerability will be detected or that the code will be entirely secure against all possible attacks. The assessment is conducted based on the specific commit and version of the code provided. Any subsequent modifications to the code may introduce new vulnerabilities that were absent during the initial review. Therefore, any changes made to the code require a new security review to ensure that the code remains secure. Please be advised that the Cantina Managed security review is not a replacement for continuous security measures such as penetration testing, vulnerability scanning, and regular code reviews.

1.3 Risk assessment

Severity level	Impact: High	Impact: Medium	Impact: Low
Likelihood: high	Critical	High	Medium
Likelihood: medium	High	Medium	Low
Likelihood: low	Medium	Low	Low

1.3.1 Severity Classification

The severity of security issues found during the security review is categorized based on the above table. Critical findings have a high likelihood of being exploited and must be addressed immediately. High findings are almost certain to occur, easy to perform, or not easy but highly incentivized thus must be fixed as soon as possible.

Medium findings are conditionally possible or incentivized but are still relatively likely to occur and should be addressed. Low findings are a rare combination of circumstances to exploit, or offer little to no incentive to exploit but are recommended to be addressed.

Lastly, some findings might represent objective improvements that should be addressed but do not impact the project's overall security (Gas and Informational findings).

2 Security Review Summary

Sky Protocol is a decentralised protocol developed around the USDS stablecoin.

From Feb 4th to Feb 7th the Cantina team conducted a review of [sparklend-advanced](#) on commit hash [fffb9b5b](#) (tag [v1.6.0-beta.0](#)). The team identified a total of **1** issues:

Issues Found

Severity	Count	Fixed	Acknowledged
Critical Risk	0	0	0
High Risk	0	0	0
Medium Risk	0	0	0
Low Risk	1	1	0
Gas Optimizations	0	0	0
Informational	0	0	0
Total	1	1	0

The Cantina Managed team reviewed Sky's [sparklend-advanced](#) holistically on commit hash [e2ba58d0](#) (tag [v1.6.0](#)) and concluded that all findings were addressed and no new vulnerabilities were identified.

3 Findings

3.1 Low Risk

3.1.1 Exchange-rate adapters lack staleness checks against upstream pause/cached update edge cases

Severity: Low Risk

Context: [RSETHExchangeRateOracle.sol#L31-L40](#)

Description: The REETH adapter builds a USD price by multiplying two external values, but it does not check whether either value is old. For rsETH, Spark reads a cached value (`rsETHPrice`) from Kelp's oracle. That oracle can pause on large downside moves and stop updating the cached value (e.g. [LRTOracle.sol#L114-L120](#)). During that pause, Spark can keep using an old price. When updates resume, the price can jump in one step.

If this asset is borrowable, this stale-then-jump behavior is directly dangerous for users: a delayed repricing can make borrow debt jump suddenly and push positions from healthy to liquidatable in one update.

Recommendation: There is not a simple fix other than closely monitoring and document this risk that comes with these oracles.

Sky: Documented this for the ratio style oracles in commit [17d675a2](#). Also for the rsETH exchange rate oracle we are not using this for any borrowable asset.

Cantina Managed: Fix verified.